MUHAMMAD YOUNIS

Roll No: F23BINFT1M01059

MACHINE LEARNING-BASED PAKISTANI USED CAR PRICE PREDICTION SYSTEM

Final Year Project (FYP) — Project Proposal

Student Name	Muhammad Younis
Roll Number	F23BINFT1M01059
Project Domain	Machine Learning / Web Application

1. Project Overview

The proposed project is a web-based system that estimates the market price of used cars in Pakistan using Machine Learning. A user will enter a car's details—such as make, model, variant, manufacturing year, engine capacity, mileage, fuel type, transmission, city and overall condition—and the system will provide an estimated price based on patterns learned from a Pakistani used-car dataset. The aim is to make price checking easier and help users make more informed buying and selling decisions.

2. Problem Statement

Used-car prices can vary significantly depending on the model, year, mileage, location and condition. Buyers and sellers may find it difficult to judge whether an asking price is reasonable. This project will use historical/listing data and a trained Machine Learning model to generate a consistent estimated price. The prediction will be an estimate, not a guaranteed sale price, and its reliability will depend on the quality and coverage of the dataset.

3. Main Features

- **Car price prediction:** Estimate a used car's price from its specifications and market-related details.
- **Pakistani car coverage:** Support car makes and models found in the selected Pakistan-focused dataset.
- **Condition-aware estimate:** Include user-provided condition information as an input to the model.
- **Image-based condition scan (planned feature):** A scan feature has been started/prototyped to examine a car image for visible condition cues and may help refine the estimate. This feature is still under development and is not yet complete; its accuracy and scope will need testing.
- **Simple web interface:** Provide an easy form for entering car details and viewing the predicted price.

4. Proposed Workflow

1. Collect and prepare Pakistani used-car listing data, then clean missing or inconsistent values.
2. Explore the dataset and select useful features such as model, year, mileage, engine capacity and city.
3. Train and evaluate suitable regression models; compare performance using appropriate metrics such as MAE and RMSE.
4. Build a web interface where users enter vehicle details and receive an estimated price.
5. Integrate and test the image-based condition scan as an advanced feature if feasible; clearly communicate its limitations.

5. Proposed Technologies

Area	Technology
Frontend	HTML5, CSS3, JavaScript, React.js, Tailwind CSS
Backend	Node.js, Express.js, REST API
Machine Learning	Python, Pandas, NumPy, Scikit-learn
Data visualization / analysis	Matplotlib, Seaborn, Jupyter Notebook / Google Colab
Database	MongoDB
Image scan (planned)	Python-based image processing / computer vision; final approach to be selected during development
Development tools	VS Code, Git, GitHub, Postman

6. Expected Outcome

The expected outcome is a functional web application that predicts used-car prices from supported vehicle details using a Machine Learning model trained on Pakistani data. The project will also explore an image-based condition scan as an additional feature. Model performance will be evaluated on held-out data, and predictions will be presented as estimates rather than exact market guarantees.

7. Project Scope Note

The final list of supported cars and prediction quality will depend on the dataset collected and prepared. The image-based condition scan is a planned enhancement that requires further development and testing; it should not be considered a completed or fully accurate feature at the proposal stage.